UP Board Class 12 Biology - 348 (CL) 2023 Question Paper with Solutions

Time Allowed :3 Hours	**Maximum Marks :**100	**Total Questions :**9

General Instructions

Read the following instructions very carefully and strictly follow them:

1. There are in all nine questions in this question paper.
2. All questions are compulsory.
3. In the beginning of each question, the number of parts to be attempted are clearly mentioned.
4. Marks allotted to the questions are indicated against them.
5. Start solving from the first question and proceed to solve till the last one. Do not waste your time over a question you cannot solve.

Multiple Choice Questions

1.(a) Which of the following is not a characteristic of pollination by wind?

(A) Light pollen grain
(B) Well exposed stamen
(C) Large feathery stigma
(D) Multiple ovule

Correct Answer: (D) Multiple ovule

Solution:
Step 1: Understanding wind pollination (anemophily).
Wind-pollinated plants rely on air currents to carry pollen from the anther to the stigma. To facilitate this process, specific structural adaptations are present in their flowers.

Step 2: Key characteristics of wind-pollinated flowers.

- **Light and non-sticky pollen grains:** Easily carried by the wind.
- **Well-exposed stamens:** To allow pollen grains to be released freely into the air.
- **Large, feathery stigmas:** Increase the surface area for capturing airborne pollen.

Step 3: Elimination of the odd feature.

- **(A) Light pollen grain** – Feature of wind pollination
- **(B) Well exposed stamen** – Feature of wind pollination
- **(C) Large feathery stigma** – Feature of wind pollination
- **(D) Multiple ovule** – Not specifically related to wind pollination

Step 4: Conclusion.
The correct answer is:

$$\boxed{\text{(D) Multiple ovule}}$$

Quick Tip

Wind-pollinated flowers are small, inconspicuous, and produce large quantities of light pollen grains with exposed stamens and feathery stigmas.

(b) Sickle cell anemia is caused by substitution of Glutamic Acid by which amino acid?

(A) Lysine (Lys)
(B) Methionine (Met)
(C) Valine (Val)
(D) Serine (Ser)

Correct Answer: (C) Valine (Val)

Solution:
Step 1: Understanding sickle cell anemia.
Sickle cell anemia is a genetic disorder caused by a point mutation in the gene that codes for the **beta-globin chain** of hemoglobin (Hb). This mutation leads to the formation of an abnormal hemoglobin variant known as **HbS**.

Step 2: Nature of the mutation.
The mutation occurs at the **sixth codon** of the beta-globin gene, where the nucleotide triplet `GAG` (which codes for **Glutamic acid**) is replaced by `GTG` (which codes for **Valine**).

Step 3: Biochemical consequence.
This single amino acid substitution — from hydrophilic **Glutamic acid** to hydrophobic **Valine** — alters the structure of hemoglobin, especially under low oxygen tension. This causes hemoglobin to polymerize, leading red blood cells to take on a sickle shape. These sickled cells are rigid and prone to blocking capillaries, causing pain, anemia, and organ damage.

Step 4: Conclusion.
Hence, the correct amino acid replacing Glutamic acid in sickle cell anemia is:

$$\boxed{\text{(C) Valine}}$$

> Quick Tip
>
> In sickle cell anemia, Glutamic Acid is replaced by Valine at the 6th position of the beta-globin chain of hemoglobin, leading to sickling of RBCs.

(c) How many peptide bonds does each antibody molecule have?

(A) Six
(B) Four
(C) Two
(D) One

Correct Answer: (B) Four

Solution:
Step 1: Understanding antibody structure.
An antibody (immunoglobulin) is a Y-shaped protein composed of:

- Two identical **heavy chains**
- Two identical **light chains**

These chains are held together and stabilized by **disulfide bonds**, not peptide bonds (peptide bonds form the linear amino acid chains themselves).

Step 2: Disulfide bond arrangement.
- One disulfide bond links each **light chain to a heavy chain** (2 bonds).
- Two disulfide bonds connect the **two heavy chains** (2 bonds).

Thus, there are typically **four interchain disulfide bonds** in a basic antibody (IgG) structure.

Step 3: Conclusion.
The correct answer is:

$$\boxed{\text{(B) Four}}$$

(Note: These are **disulfide bonds**, not peptide bonds.)

> Quick Tip
>
> Antibodies are Y-shaped proteins with two heavy and two light chains connected by four peptide (disulfide) bonds.

(d) Which of the following condition is not maintained in a Bioreactor for optimum growth condition?

(A) Temperature
(B) Vitamin
(C) Salts
(D) None of the above

Correct Answer: (B) Vitamin

Solution:
Step 1: Understanding bioreactors.
A bioreactor is a controlled environment designed to support the large-scale growth of microorganisms or cells. It ensures the maintenance of optimal physical and chemical conditions such as temperature, pH, oxygen levels, and essential nutrients.

Step 2: Analysis of options.
- (A) Temperature: Actively regulated in bioreactors to promote optimal growth of cells or microbes.
- (B) Vitamin: Not commonly monitored or controlled; many microorganisms can synthesize required vitamins.
- (C) Salts: Crucial for maintaining ionic balance and nutrient availability; typically managed in the system.
- (D) None of the above: Incorrect, since vitamins are not directly regulated in bioreactors.

Step 3: Conclusion.
The correct answer is (B) Vitamin, as it is not specifically regulated in bioreactor systems.

Quick Tip

Bioreactors maintain conditions like temperature, pH, oxygen supply, and salts, but vitamins are usually not externally regulated.

Subjective Questions

2.(a) Oxalis and Commelina produce which two types of flowers? Name them.

Solution:
Plants like *Oxalis* and *Commelina* produce two distinct types of flowers:

1. **Chasmogamous flowers:** - These flowers are open, with exposed anthers and stigma.
   - They facilitate both cross-pollination and self-pollination.
2. **Cleistogamous flowers:** - These flowers remain closed, with anthers and stigma enclosed within.
   - They promote obligate self-pollination and ensure seed formation even in the absence of external pollinators.

Conclusion:
The presence of both chasmogamous and cleistogamous flowers in plants like *Oxalis* and *Commelina* provides a reliable reproductive strategy across varying environmental conditions.

Quick Tip

Chasmogamous = open flowers (cross/self-pollination). Cleistogamous = closed flowers (only self-pollination).

(b) Which bacterium did Frederick Griffith use for his research? Also mention the name of the disease caused by that bacterium.

Solution:
Frederick Griffith's Experiment: In 1928, Frederick Griffith conducted his famous experiment on bacterial transformation.
Bacterium Used: - He worked with **Streptococcus pneumoniae**, a bacterium that causes pneumonia in humans and animals.
Details: - Griffith studied two strains of the bacterium:

- **S-strain (Smooth):** Virulent, disease-causing strain with a polysaccharide capsule.
- **R-strain (Rough):** Non-virulent, harmless strain without capsule.

- His experiment showed that some "transforming principle" from the heat-killed S-strain bacteria converted the R-strain into a virulent form. - This laid the foundation for the discovery of DNA as the genetic material.
Disease Caused: Pneumonia.

Quick Tip

Griffith's experiment with *Streptococcus pneumoniae* demonstrated transformation, proving that genetic material can be transferred between organisms.

(c) Which part of female reproductive system is responsible for the collection of ovum after ovulation?

Solution:

Step 1: Ovulation.
During ovulation, the ovary releases a mature ovum (egg) into the peritoneal cavity.
Step 2: Structure responsible.
The **funnel-shaped part of the fallopian tube, called the infundibulum**, is responsible for collecting the ovum.

- The margins of the infundibulum have finger-like projections called **fimbriae**.
- Fimbriae are close to the ovary and help in guiding the ovum into the fallopian tube.

Step 3: Conclusion.
Thus, the **infundibulum with fimbriae of the fallopian tube** collects the ovum after ovulation.

Quick Tip

Remember: Ovum → collected by fimbriae of infundibulum → passes into fallopian tube → fertilization site.

(d) Write any two reasons why Mendel's theory of inheritance remained unrecognised till 1900.

Solution:

Step 1: Background.
Mendel published his work on pea plant experiments in 1865, but it remained unnoticed until 1900, when rediscovered independently by De Vries, Correns, and Tschermak.
Step 2: Reasons for unrecognition.

1. **Ahead of time:** Mendel's work was conceptually advanced, and the scientific community of his time could not appreciate the significance of statistical analysis and probability in biology.
2. **Lack of knowledge of chromosomes:** Cytological studies of chromosomes and their role in heredity were not understood then, so Mendel's "factors" (genes) could not be correlated with physical structures.

Step 3: Conclusion.
Only after cytology and genetics advanced, Mendel's laws were recognized as the foundation of heredity.

Quick Tip

Mendel's theory was ignored due to lack of cytological proof and advanced concepts. Rediscovered in 1900 by three scientists.

(e) Write the names of any two cyanobacteria responsible for nitrogen fixation.

Solution:
Cyanobacteria, also known as **blue-green algae**, are photosynthetic prokaryotes that can fix atmospheric nitrogen into usable forms (like ammonia), enriching soil fertility. Some of them possess specialized cells called **heterocysts** where nitrogen fixation occurs.
Examples:

1. **Nostoc**
2. **Anabaena**

Note: Both Nostoc and Anabaena are commonly found in rice fields and play an important role as biofertilizers in agriculture.

> Quick Tip
>
> Remember: Nostoc and Anabaena are the two most common nitrogen-fixing cyanobacteria used as natural biofertilizers in paddy fields.

3.(a) Which type of cross is used to determine the genotype of tall pea plant? Explain.

Solution:
The type of cross used to determine the genotype of a tall pea plant is called a **Test Cross**.
Explanation: - A tall pea plant may have either of the two genotypes:

- Homozygous dominant (TT)
- Heterozygous (Tt)

- To find out whether the tall plant is TT or Tt, it is crossed with a homozygous recessive dwarf plant (tt).
Case 1: If the tall plant is TT (homozygous)

$$TT \times tt \;\; \Rightarrow \;\; 100\%\, Tall(Tt)$$

All offspring will be tall.
Case 2: If the tall plant is Tt (heterozygous)

$$Tt \times tt \;\; \Rightarrow \;\; 50\%\, Tall(Tt) \;\&\; 50\%\, Dwarf(tt)$$

Conclusion: - If all progeny are tall → the parent tall plant is homozygous (TT). - If progeny are 50

Quick Tip

A test cross is always done with a recessive parent to determine the unknown genotype of the dominant phenotype.

(b) Explain Co-extinctions.

Solution:
Definition: Co-extinction refers to the phenomenon in which the extinction of one species leads to the extinction of another species that is closely associated with it. It usually happens when two species are interdependent for survival.
Explanation: - Many species exist in ecological relationships such as host–parasite, plant–pollinator, and predator–prey. - If one species becomes extinct, the dependent species also faces extinction because it cannot survive without the other.
Examples:

- If a host fish species becomes extinct, its specific parasites also get extinct.
- If a flowering plant species goes extinct, its specialized pollinator (like a particular bee or butterfly) will also vanish.

Conclusion: Co-extinction highlights the interconnectedness of ecosystems, showing that the loss of one species can trigger a chain reaction of extinctions.

Quick Tip

"Extinction of one species leads to the extinction of another dependent species" $\rightarrow$ This is co-extinction.

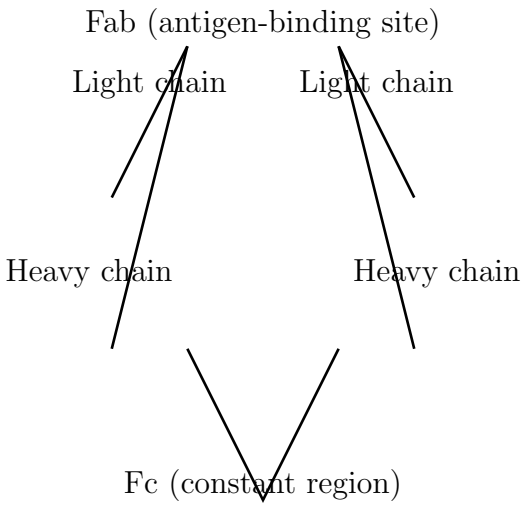
(c) Draw only the structure of antibody molecule.

Solution:

The antibody (immunoglobulin) is a Y-shaped glycoprotein molecule made up of:

- Two identical heavy (H) chains.
- Two identical light (L) chains.
- Variable region at the tips of the arms for antigen binding.
- Constant region forming the stem of the Y.

Diagram:



Quick Tip

Antibody = Y-shaped with 2 heavy + 2 light chains. Fab = antigen binding, Fc = constant region.

(d) In *lac* operon, 'Z' and 'Y' genes code for which enzymes? Also mention their functions.

Solution:

Step 1: Background.
The *lac* operon in *E. coli* regulates lactose metabolism. It has structural genes (Z, Y, A) controlled by a promoter, operator, and regulator gene.
Step 2: Role of Z and Y genes.

- **Z gene:** Codes for **-galactosidase enzyme**.
  - Function: Hydrolyses lactose into glucose and galactose.
- **Y gene:** Codes for **Permease enzyme**.
  - Function: Increases permeability of the cell membrane to allow entry of lactose into the bacterial cell.

Step 3: Conclusion.
Thus, in the *lac* operon, Z and Y genes play key roles in lactose utilization: Z (breakdown) and Y (transport).

Quick Tip

Z = -galactosidase (breaks lactose), Y = Permease (transports lactose).

(e) Comment on Organic Farming.

Solution:
Organic farming is a sustainable agricultural practice that avoids the use of synthetic fertilizers, pesticides, genetically modified organisms (GMOs), and growth regulators. Instead, it relies on natural processes and eco-friendly techniques to maintain soil fertility and control pests.
Key Features of Organic Farming:

- Use of organic manures (compost, farmyard manure, green manure) to maintain soil fertility.
- Biological pest control using natural predators, parasites, and biopesticides like neem extract.
- Crop rotation and intercropping to maintain soil health and prevent pest build-up.
- Integration of livestock and crop production for recycling nutrients.

Advantages:

- Produces healthier and chemical-free food.
- Protects the environment by reducing soil and water pollution.
- Maintains long-term soil fertility and promotes biodiversity.

Limitations:

- Lower yield compared to modern intensive farming.
- Requires more labor and careful management.
- Organic products are often costlier.

Conclusion: Organic farming is eco-friendly and sustainable, making it important for future food security and environmental conservation, even though it may give comparatively lower yields.

Quick Tip

Organic farming = eco-friendly farming. No chemicals, only natural manures and biological methods, ensuring sustainable agriculture.

4.(a) **What is the difference between false fruit and true fruit? Under which category does the apple fall and why?**

Solution:
True Fruit: - A true fruit develops only from the ovary of the flower after fertilization. - Example: Mango, Tomato, Grapes.
False Fruit: - A false fruit develops not only from the ovary but also from other floral parts such as thalamus, receptacle, or perianth. - Example: Apple, Strawberry, Cashew.
Apple Category: - The apple is a **false fruit** because its edible fleshy part is formed from the thalamus (receptacle) and not just the ovary.
Conclusion: Thus, the apple falls under false fruits due to its origin from floral parts in addition to the ovary.

Quick Tip

True fruits = ovary only; False fruits = ovary + other floral parts. Apple = false fruit.

(b) Why did Mendel select pea plants for his experiments?

Solution:
Gregor Johann Mendel, the father of genetics, selected **pea plants (Pisum sativum)** for his hybridization experiments due to the following reasons:

1. **Distinct contrasting traits:** Pea plants show easily observable contrasting traits such as tall/dwarf plants, round/wrinkled seeds, green/yellow pods, etc.
2. **Short life cycle:** They have a short life span, which allowed Mendel to study several generations in a short period.
3. **Ease of cultivation:** Pea plants are easy to grow and require simple care.
4. **Natural self-pollination:** Ensures genetic stability.
5. **Artificial cross-pollination possible:** Flowers are large and bisexual, making artificial hybridization (manual crossing) easy.

Conclusion: Because of these advantages, Mendel obtained reliable and reproducible results, which helped him formulate the **Laws of Inheritance**.

Quick Tip

Mendel chose pea plants due to their short life span, distinct traits, and ability for both self- and cross-pollination.

(c) Explain Stirred-tank Reactor.

Solution:

Step 1: Definition.
A Stirred-tank Reactor (STR) is a common type of bioreactor used in industrial microbiology and biotechnology for large-scale production of microbial products, enzymes, antibiotics, and recombinant proteins.
Step 2: Structure.

- It is a cylindrical vessel made of stainless steel.
- Equipped with:
  - **Agitator (impeller):** Provides mixing of culture medium.
  - **Sparger:** Supplies sterile air (oxygen).
  - **Baffles:** Prevent vortex formation and improve aeration.
  - **Cooling jacket:** Maintains optimum temperature.

Step 3: Working.

- Microbes are inoculated in nutrient medium inside the reactor.
- Continuous stirring ensures uniform distribution of nutrients and oxygen.
- Waste gases are released through an outlet, while product is harvested from the vessel.

Step 4: Applications.

- Production of antibiotics (penicillin, streptomycin).
- Production of alcohols, enzymes, vaccines, recombinant proteins (like insulin).

> Quick Tip
>
> Stirred-tank reactor = mixing + aeration = uniform environment for microbial growth and product yield.

(d) Explain Logistic Growth with equation.

Solution:

Step 1: Definition.
Logistic growth describes population growth when resources are limited. Initially, population grows rapidly (exponential), but later slows down and stabilizes at a maximum value called the **carrying capacity (K)**.
Step 2: Logistic growth curve.

- Shape: S-shaped (sigmoid curve).

- Phases:
  1. Lag phase – slow growth.
  2. Exponential phase – rapid growth.
  3. Deceleration phase – growth slows due to competition.
  4. Stationary phase – population stabilizes at carrying capacity.

Step 3: Logistic growth equation.

$$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$$

Where:

- $\frac{dN}{dt}$ = rate of population growth.
- r = intrinsic rate of natural increase.
- N = population size at time t.
- K = carrying capacity of the environment.

Step 4: Conclusion.
Logistic growth is more realistic than exponential growth as it considers limited resources and carrying capacity.

> Quick Tip
>
> Exponential = J-shaped (unlimited resources). Logistic = S-shaped (limited resources, carrying capacity).

5.(a) Draw the schematic chart representation of Oogenesis.

Solution:
Oogenesis is the process of formation of mature female gametes (ova) from oogonia in the ovary. It begins during the embryonic stage and is completed after puberty in females.
Schematic Chart of Oogenesis:

$$\text{Oogonium (2n)} \xrightarrow{\text{Mitosis}} \text{Primary Oocyte (2n)} \xrightarrow{\text{Meiosis I}} \text{Secondary Oocyte (n)+First Polar Body (n)} \xrightarrow{\text{Me}}$$

Key Points:

- Primary oocytes remain arrested in prophase I until puberty.
- Secondary oocyte is released at ovulation and remains arrested at metaphase II.
- Meiosis II is completed only after fertilization.

Quick Tip

Oogenesis → Oogonium → Primary oocyte → Secondary oocyte (+ polar body) → Ovum (after fertilization).

(b) Write short notes on Active Immunity and Vaccination.

Solution:

Active Immunity:

- Active immunity is the immunity developed by the body when it is exposed to antigens, either naturally (through infection) or artificially (through vaccines).
- The immune system produces antibodies and memory cells, providing long-lasting protection.
- Example: Recovery from measles infection or immunity after polio vaccination.

Vaccination:

- Vaccination is an artificial method of providing active immunity.
- A vaccine is a preparation containing dead or weakened pathogens, or their antigenic parts, which stimulate the immune system without causing disease.
- Vaccines help the body produce memory cells, so that during actual infection, the immune response is faster and stronger.
- Examples: Polio vaccine, BCG vaccine (for tuberculosis), Hepatitis B vaccine.

Conclusion: Vaccination is one of the most effective public health measures to prevent infectious diseases and is based on the principle of inducing active immunity.

Quick Tip

Active immunity = long-term defense produced by body's own antibodies. Vaccination = safe exposure to antigens → stimulates active immunity and memory cell formation.

(c) Mention any three vectorless methods that are used to introduce recombinant DNA into competent host cell.

Solution:
Vectorless methods are techniques in which recombinant DNA is directly introduced into the host cell without using a carrier (vector). Three important methods are:

1. **Microinjection:** - Direct injection of recombinant DNA into the nucleus of an animal or plant cell using a fine needle. - Commonly used in animal cells, especially in producing transgenic organisms.
2. **Biolistics (Gene Gun Method):** - DNA is coated onto microscopic particles of gold or tungsten. - These particles are shot at high velocity into plant cells. - Successfully used for introducing genes into plants like maize, rice, and wheat.
3. **Electroporation:** - Host cells are exposed to brief electrical pulses that create transient pores in the cell membrane. - Through these pores, recombinant DNA enters the cell. - Widely used for bacteria, yeast, and animal cells.

Conclusion: Vectorless methods are powerful alternatives to vector-based transfer, especially in plants and animals, for creating genetically modified organisms.

Quick Tip

Remember: Microinjection → nucleus, Gene gun → plants, Electroporation → temporary pores by electric pulses.

(d) Explain how distribution of animals and plants is affected by light, with examples.

Solution:
Effect of Light on Plants: - Light is one of the most important abiotic factors influencing plant distribution. - Plants show responses to both intensity and duration of light (photoperiodism).

- High light intensity favors growth of sun-loving plants (heliophytes) such as sunflower.
- Shade-loving plants (sciophytes) like ferns thrive in low light intensity.
- Flowering in plants like wheat and rice depends on day length (short-day and long-day plants).

Effect of Light on Animals: - Many animal activities like migration, reproduction, feeding, and diurnal rhythms are controlled by light.

- Birds such as migratory geese and Siberian cranes use changes in day length as cues for migration.
- Reproduction in many animals is photoperiod-dependent (e.g., sheep breed in short-day conditions).
- Nocturnal animals like owls and bats are active in darkness and rest during the day.

Conclusion: Light governs photosynthesis in plants and influences behavior, physiology, and distribution of animals. Thus, it plays a central role in shaping ecosystems.

Quick Tip

Plants depend on light for photosynthesis and flowering (photoperiodism), while animals rely on light for migration, reproduction, and daily activity cycles.

6.(a) A colourblind father has a daughter with normal vision and the daughter marries a man with normal vision. What is the probability of her son being a carrier for colourblindness? Explain with pedigree chart.

Solution:

Step 1: Background.
Colourblindness is an X-linked recessive disorder.

- Females have two X chromosomes (XX).
- Males have one X and one Y chromosome (XY).
- Since males have only one X chromosome, if it carries the defective allele, they are colourblind.
- Females become carriers when they have one defective allele and one normal allele.

Step 2: Cross explanation.

- Father = X^cY (colourblind).
- Mother (normal, not a carrier) = X^NX^N.
- Daughter inherits X^c from father and X^N from mother = X^NX^c (carrier, normal vision).
- Daughter marries a normal man (X^NY).

Step 3: Punnett square.

	X^N	Y
X^N	X^NX^N	X^NY
X^c	X^NX^c	X^cY

Step 4: Results.

- Daughters: 50% normal (X^NX^N), 50% carriers (X^NX^c).
- Sons: 50% normal (X^NY), 50% colourblind (X^cY).

Step 5: Answer.
Probability of her son being a carrier = 0, because males cannot be carriers (they have only one X chromosome). A son will either be **normal (50%)** or **colourblind (50%)**.

Quick Tip

Males cannot be carriers for X-linked traits; only females can. Sons are either normal or affected.

(b) Mention any three drawbacks of insulin obtained from the pancreas of slaughtered cattle and pigs.

Solution:

Earlier, insulin for diabetic patients was extracted from the pancreas of cattle and pigs. However, it had several drawbacks:

1. **Immunological reactions:** Animal insulin differed slightly in amino acid sequence from human insulin, sometimes causing allergic reactions or immune responses in patients.
2. **Risk of infections:** There was a possibility of transmitting animal-borne pathogens and contaminants through animal-derived insulin.
3. **Limited supply and cost:** Large numbers of animals had to be slaughtered to obtain small amounts of insulin, making the process costly and unsustainable.

Conclusion.
These drawbacks were overcome by developing **recombinant human insulin** using genetically engineered *E. coli*, which is safer, cheaper, and identical to natural human insulin.

Quick Tip

Animal insulin = risk of allergy, infection, and costly. Recombinant human insulin = safe and identical to human.

(c) Comment on *in-situ* and *ex-situ* conservation.

Solution:
Biodiversity conservation strategies are of two main types: **in-situ** (on-site) and **ex-situ** (off-site).
1. In-situ Conservation (On-site):

- Conservation of species in their natural habitats.
- Protects entire ecosystems and natural processes.
- Examples: Biosphere reserves (Nilgiri, Nanda Devi), National parks (Kaziranga, Gir), Wildlife sanctuaries.

- **Advantage:** Maintains natural evolutionary processes and preserves interactions among species.

2. Ex-situ Conservation (Off-site):

- Conservation of species outside their natural habitats.
- Used when species are critically endangered or natural habitats are lost.
- Examples: Botanical gardens, zoological parks, seed banks (e.g., Svalbard Global Seed Vault), cryopreservation of gametes.
- **Advantage:** Provides intensive care and protection to endangered species.

Conclusion: Both in-situ and ex-situ methods are complementary. In-situ ensures natural conservation, while ex-situ serves as backup for threatened species.

Quick Tip

In-situ = conserve in natural habitat (National Parks). Ex-situ = conserve outside habitat (Seed banks, Zoos).

(d) Explain Competition and Parasitism with examples.

Solution:

Competition:

- An interaction where individuals of the same or different species compete for limited resources such as food, light, water, or space.
- Competition can be **interspecific** (between species) or **intraspecific** (within species).
- Example: Flamingos and fishes competing for zooplankton in lakes; two plants competing for sunlight in a dense forest.

Parasitism:

- A relationship where one organism (parasite) benefits at the expense of another organism (host).
- The parasite derives food and shelter, while the host is harmed.
- Example: Plasmodium (malarial parasite) in humans; Cuscuta (dodder plant) parasitizing other plants.

Difference:

- Competition = both species are harmed (, interaction).
- Parasitism = parasite benefits, host harmed (+, interaction).

Quick Tip

Competition = struggle for resources (both harmed). Parasitism = one benefits (parasite), other harmed (host).

---

7. Differentiate between any two of the following:

(i) Hypocotyl and Epicotyl
(ii) Perisperm and Pericarp
(iii) Integument and Testa

Solution:

(i) Difference between Hypocotyl and Epicotyl:

Hypocotyl	**Epicotyl**
Part of the embryonic axis below the cotyledons.	Part of the embryonic axis above the cotyledons.
Develops into the radicle (root system).	Develops into the plumule (shoot system).
Example: Elongates during germination in dicots like bean.	Example: Gives rise to stem and leaves.

(ii) Difference between Perisperm and Pericarp:

Perisperm	**Pericarp**
Remnant of nucellus present in some seeds.	Wall of the ovary developed after fertilization.
It is nutritive in function.	It forms protective layers of fruit (epicarp, mesocarp, endocarp).
Example: Black pepper, beet.	Example: Mango, pea, tomato.

(iii) Difference between Integument and Testa:

Integument	**Testa**
Protective covering of ovule before fertilization.	Seed coat formed from integument after fertilization.
Part of ovule structure.	Part of mature seed structure.

Quick Tip

Hypocotyl → root part, Epicotyl → shoot part. Perisperm → nucellus, Pericarp → ovary wall. Integument → ovule covering, Testa → seed coat.

OR
Define Parturition. Give an account of different hormones involved during this

process.

Solution:
Definition: Parturition is the process of expelling the fully developed fetus from the mother's uterus at the end of gestation (childbirth).
Hormonal Control of Parturition:

1. **Oxytocin:** Secreted by the posterior pituitary. Stimulates strong uterine contractions during labor.
2. **Relaxin:** Relaxes the ligaments of the pelvic region and softens the cervix for easy passage of the baby.
3. **Estrogen and Progesterone:** During pregnancy, progesterone inhibits uterine contractions. Near parturition, progesterone decreases while estrogen increases, stimulating uterine contractions.
4. **Prostaglandins:** Enhance contraction of uterine muscles and work synergistically with oxytocin.

Conclusion: Thus, parturition is a neuro-endocrine mechanism regulated by maternal and fetal signals involving hormones like oxytocin, estrogen, progesterone, relaxin, and prostaglandins.

Quick Tip

Progesterone maintains pregnancy, but a drop in its level along with rise in oxytocin and prostaglandins triggers parturition.

---

8. Describe the disorders from the following chromosomal abnormal arrangements: (i) Trisomy of Chromosome 21 (ii) XXY

Solution:

(i) Trisomy of Chromosome 21 (Down's Syndrome):

- Caused by the presence of an extra copy of chromosome 21 (2n = 47).
- The condition arises due to non-disjunction of chromosome 21 during meiosis.
- **Symptoms:**
  - Mental retardation and delayed cognitive development.
  - Short stature, round face, slanting eyes, and flat nasal bridge.
  - Heart defects and poor muscle tone.
- More frequent in children of older mothers (¿35 years).

(ii) XXY (Klinefelter's Syndrome):

- Caused by the presence of an extra X chromosome in males (47, XXY).
- Results from non-disjunction of sex chromosomes during gamete formation.
- **Symptoms:**
  - Male individuals with tall stature and long limbs.
  - Underdeveloped testes leading to sterility.
  - Enlarged breasts (gynecomastia) and sparse facial/body hair.
  - Reduced testosterone levels.

Conclusion:
Both conditions are chromosomal disorders caused by abnormal segregation of chromosomes during meiosis. They affect physical, mental, and reproductive health.

> Quick Tip
>
> Down's = Trisomy 21 (47 chromosomes). Klinefelter's = XXY (47 chromosomes, male sterility).

OR

Explain the salient features of Genetic code.

Solution:

The genetic code refers to the set of rules by which the sequence of nucleotides in mRNA is translated into amino acids during protein synthesis.
Salient features of genetic code:

1. **Triplet code:** Each codon is a sequence of three nucleotides. Example: AUG.
2. **Universal:** Same codons specify the same amino acids in almost all organisms (exceptions: mitochondrial DNA).
3. **Non-overlapping:** Codons are read one after another without overlapping.
4. **Comma-less:** There are no punctuation marks between codons; reading is continuous.
5. **Degenerate code:** One amino acid may be coded by more than one codon (e.g., leucine has 6 codons).
6. **Unambiguous:** Each codon codes for only one amino acid.
7. **Start codon:** AUG codes for methionine and signals initiation of translation.
8. **Stop codons:** UAA, UAG, and UGA act as termination signals.

Conclusion:
The genetic code is a highly precise, universal, and conserved system that ensures accurate translation of genetic information into proteins.

> Quick Tip
>
> Genetic code = triplet, universal, degenerate, unambiguous, continuous, with start (AUG) and stop codons (UAA, UAG, UGA).

---

9. Write short notes on any two of the following:
(i) Adenosine Deaminase Deficiency (ii) Interspecific Competition with example (iii) Human Genome Project

Solution:

(i) Adenosine Deaminase (ADA) Deficiency:

- ADA deficiency is a rare genetic disorder that causes severe combined immunodeficiency (SCID).
- It is caused by the mutation in the gene encoding the enzyme adenosine deaminase, which is essential for purine metabolism.
- Lack of ADA leads to accumulation of toxic metabolites, resulting in the destruction of lymphocytes (T-cells, B-cells), leading to a weak immune system.
- Patients become highly susceptible to infections.
- Treatment: Enzyme replacement therapy, bone marrow transplantation, and gene therapy.

(ii) Interspecific Competition (with example):

- Interspecific competition occurs when individuals of different species compete for the same limited resources such as food, water, or space.
- It is a $(-,-)$ interaction where both species are negatively affected.
- Example: Flamingos and fishes competing for zooplankton in lakes. Both species have reduced access to food.
- Another Example: Paramecium aurelia and Paramecium caudatum grown together compete for food; one species outcompetes the other (Gause's competitive exclusion principle).

(iii) Human Genome Project (HGP):

- Launched in 1990 and completed in 2003, HGP was an international research program aimed at mapping and sequencing the entire human genome.
- Objectives: Identify all genes in human DNA, determine base pair sequences, and store the data for further biological studies.

- Outcomes: About 20,000–25,000 human genes were identified. The project provided insights into genetic diseases, molecular medicine, and evolutionary biology.
- Applications: Development of new drugs, better diagnosis of genetic disorders, and personalized medicine.

> Quick Tip
>
> ADA deficiency = immune disorder; Interspecific competition = struggle between species; HGP = sequencing human genome.

OR
What is contact inhibition? Explain how the normal cell transforms into cancerous cell.

Solution:

Contact Inhibition:

- Contact inhibition is the property of normal cells to stop dividing when they come into contact with neighboring cells.
- It ensures controlled growth and maintains tissue architecture.

Transformation into Cancerous Cells:

- In cancerous cells, contact inhibition is lost. They continue to divide uncontrollably even after touching adjacent cells.
- This uncontrolled proliferation leads to the formation of a mass of cells called a tumor.
- Mutations in proto-oncogenes or tumor suppressor genes disrupt normal cell cycle regulation.
- Tumors may remain benign or become malignant, invading other tissues (metastasis).

> Quick Tip
>
> Normal cells stop dividing on contact (contact inhibition). Loss of this property leads to uncontrolled growth $\rightarrow$ cancer.
